

Scalability & Future Plan

Date	24 June 2026
Team ID	SWTID-2026-3249
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how OptiCrop can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Runs on Flask's single-threaded development server	Limits concurrent request handling under real-world traffic	High
2	No live weather or IoT sensor integration	Recommendations rely on manually entered values instead of real-time field data	Medium
3	Model trained on a fixed, balanced dataset of 22 crops	Cannot recommend crops outside the trained classes	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-threaded local Flask server	Deploy with Gunicorn + Nginx and horizontal scaling behind a load balancer
2	Data Storage	Static CSV file (Crop_recommendation.csv)	Migrate to a managed database for versioned, growable training data
3	Performance	In-memory model loaded once at startup	Add caching and asynchronous request handling for high concurrency
4	Security	No authentication or input rate-limiting	Add API authentication, HTTPS, and request throttling for public deployment

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	IoT sensor integration for automatic soil data collection	3-6 months	Removes manual data entry, improves accuracy
Phase 2	Real-time weather API integration	6-9 months	Adapts recommendations to live seasonal conditions
Phase 3	Cloud deployment (AWS/GCP) with CI/CD	9-12 months	Improves reliability, speed, and scalability
Phase 4	Mobile application for field-level access	12+ months	Extends reach to farmers without desktop access